Machine Learning Project

Evaluation and improvement of probability calibration in binary classification models

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Chapter 1

Introduction

In **supervised classification problems**, the quality of a model does not depend only on its ability to correctly assign all observations to their respective classes. In many cases, it is also important that the probabilities associated with the predictions are accurate. For this reason, good predictive accuracy does not necessarily guarantee well-calibrated probabilities [1].
In fact, this project is based on determining the accuracy of these probabilities.

A classifier is **well-calibrated** when the estimated probabilities reflect the actual observed frequencies [2]. For example, for observations with an estimated 70% probability of being in a specific class, around 70% should actually be in that class.
Calibration therefore allows us to evaluate how well the accuracy of the model matches the results.

The objective of this project is therefore to evaluate and improve the calibration of two classifiers, *Logistic Regression* and *Random Forest*, on two real-world classification datasets.
The probabilities generated by the models are analysed both before and after the application of **Platt Scaling** and **Isotonic Regression**, which have been implemented from scratch.
The comparison is based on quantitative metrics and reliability diagrams, with particular attention paid to both the benefits and the limitations introduced by calibration.

Chapter 2

Theoretical Background

2.1 Probability Calibration

A **probabilistic classifier** does not just return a predicted class, but assigns to each observation a probability of belonging to the respective class.
These probabilities must be well calibrated to provide a consistent measure of the confidence of the model. In the binary case considered in this project, a model is calibrated when, *between observations that have a probability p of belonging to a class, a proportion approximately equal to p actually belongs to that class.*
Calibration is different from classification performance because a model can correctly identify many observations but still produce probabilities that do not accurately represent the observed frequencies. For this reason, classification performance alone is not sufficient.

2.2 Calibration Methods

For the reasons mentioned above, when the probabilities produced by a classifier are biased, we can apply a **post-hoc transformation**, i.e. one performed after the model has been trained.
This project considers two techniques: **Platt Scaling** and **Isotonic Regression**. In both cases, the calibration function was created using different data from that used to train the classifier, in order to prevent bias in the

probabilities.

Platt Scaling is a parametric method that transforms the classifier's output using a sigmoid function:

$$P(Y = 1 \mid f) = \frac{1}{1 + \exp(Af + B)}.$$

The parameters A and B are estimated on the calibration set. As this is a parametric and restricted method, it is more stable when there are limited data for calibration.

Isotonic Regression is a non-parametric method that uses a monotonic transformation of the scores. The project utilises the Pair-Adjacent Violators (PAV) algorithm, which combines blocks that violate the monotonicity requirement. This model offers greater flexibility, permitting the correction of more complex distortions, but increases the risk of overfitting when the calibration set is small.

2.3 Calibration Assessment

The **quality of the calibration** can be analysed both visually and quantitatively using metrics that evaluate the quality of the predicted probabilities. In this project, this information is used in conjunction with accuracy, which measures the validity of the classifications.

Reliability diagrams compare the predicted average probability with the actual proportion of positive observations for different probability intervals. A well-calibrated model produces a curve that lies close to the diagonal.

Quantitative evaluation, on the other hand, uses accuracy, Log-loss and the Brier score. **Accuracy** measures the validity of the classification, and log-loss and the Brier score evaluate the quality of the predicted probabilities. **Log-loss** penalises incorrect predictions made with a high degree of confidence more severely. The **Brier score** measures the mean squared error between the predicted probability and the observed outcome:

$$BS = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2.$$

Chapter 3

Data and Experimental Setup

3.1 Datasets

For this project, I used two real-world binary classification datasets with different characteristics.

The **Breast Cancer Wisconsin dataset**, available in *scikit-learn*, contains 569 observations and 30 numerical variables. The target variable differentiates between malignant and benign tumours, with approximately 37% being malignant and 63% benign. The dataset contains no missing values. Exploratory analysis indicates strong correlations between certain geometric characteristics of the tumour, particularly between measures relating to radius, perimeter and area.

The **Diabetes dataset**, available on *OpenML*, contains 768 observations and 8 numerical variables. The target variable differentiates between people who tested positive and those who tested negative for diabetes, with approximately 35% positive and 65% negative. The correlations between the data are more moderate: the glucose level has the strongest relationship with the target ($r = 0.47$).

3.2 Data Preprocessing

In the Diabetes dataset, exploratory analysis also revealed implausible zero values in some clinical variables. For this reason, at this stage, the implausible zeros were replaced with missing values and estimated using **MICE-style iterative imputation**.
For logistic regression, the variables were also standardised using **StandardScaler**. Imputation and standardisation were fitted exclusively to the training set and subsequently applied to the remaining data.

3.3 Data Splitting

Both datasets were split into 50% training, 30% calibration and 20% test. The **training set** is used to train the classifiers, the **calibration set** to learn the calibration transformations, and the **test set** exclusively for the final evaluation. Using a calibration set helps to guarantee that calibration is not based on the same data used to train the model.

3.4 Classification Models

Two classifiers with different characteristics were used: Logistic Regression and Random Forest.

Logistic regression is a simple parametric model that directly estimates the probability of belonging to a class via a specific relationship between the variables and the log-odds of the response.

Random Forest, on the other hand, is a non-parametric and more flexible model, based on the aggregation of predictions from multiple decision trees.

Using both makes it possible, therefore, to assess the *effect of calibration* on models with different structures.
For each model, the uncalibrated probabilities are first estimated and used as a baseline. *Platt Scaling* and *Isotonic Regression* are then applied to compare the effect of the two methods on the probabilities produced by the original models.

Chapter 4

Calibration Methods Implementation

The two calibration methods used, **Platt Scaling** and **Isotonic Regression**, were implemented from scratch using mainly NumPy. Both follow the same logic: the calibration function is trained on the scores from the calibration set and then applied to the scores from the test set.

4.1 Platt Scaling

Platt Scaling learns the two parameters of the sigmoid transformation using gradient descent. Initially, the binary targets are smoothed, replacing values 0 and 1 with slightly less extreme values to achieve better calibration. In fact, this procedure is intended to limit overfitting during calibration.
The learning is performed using the `fit()` method, which takes the scores produced by the classifier and the corresponding targets from the calibration set:`platt.fit(scores_cal, y_cal)`.

At each iteration, the probabilities are calculated using the sigmoid function and their difference from the updated target values is also calculated. This error is used to calculate the gradients of the parameters `a` e `b`.
In particular, the gradient of `a` is calculated as follows:

```
probabilities = self._sigmoid(self.a * scores + self.b)
gradient_a = np.sum((probabilities - targets) * scores) / n
```

The gradient of `b` is, of course, calculated in a similar way, and then the two parameters are updated using gradient descent.
The parameters are updated using a learning rate of 0.01. The optimisation is set to run for up to 10 000 iterations, but is terminated when the change in both parameters becomes less than the tolerance of 10^{-7}.
Having estimated the parameters on the calibration set, I apply them to the scores from the test set: `y_prob_platt = platt.predict_proba(scores_test)`. In this way, the calibration affects only the transformation of its scores into calibrated probabilities.

4.2 Isotonic Regression

Isotonic Regression was implemented using the *Pair-Adjacent Violators (PAV) algorithm*. Unlike Platt Scaling, in this case, a monotonic function in intervals is constructed directly.
In this case too, the `fit()` method takes the scores produced by the classifier and the targets from the calibration set: `isotonic.fit(scores_cal, y_cal)`.
The scores are initially sorted. Observations with identical scores are grouped together to correctly manage classifiers that produce duplicate scores. Furthermore, each unique score is associated with the number of observations and the sum of the corresponding targets.
The PAV algorithm is applied to these groups, so when two adjacent blocks violate the monotonicity condition, they are merged. In the code, this condition is checked using: `if current_value > next_value:`. The process continues until all blocks respect the monotonicity condition.
Finally, the calibration function is defined in intervals using the score thresholds (`thresholds`) and the associated probabilities (`values`).
The function is then applied to the scores in the test set: `y_prob_isotonic = isotonic.predict_proba(scores_test)`.

The main difference between the two methods therefore consists in the way the transformation is implemented: **Platt Scaling** estimates the parameters of a sigmoid function using gradient descent, and **Isotonic Regression** constructs a monotonic function in segments using progressive aggregation of blocks.

Chapter 5

Results

5.1 Breast Cancer Wisconsin Dataset

Model	Calibration	Accuracy	Log-loss	Brier
Logistic Regression	None	0.9649	0.1642	0.0319
	Platt	0.9649	**0.1434**	0.0308
	Isotonic	**0.9737**	0.9555	**0.0271**
Random Forest	None	**0.9474**	0.4345	**0.0428**
	Platt	**0.9474**	**0.1835**	0.0435
	Isotonic	0.9386	1.9140	0.0586

Table 5.1: Results on the Breast Cancer Wisconsin dataset

On the Breast Cancer dataset, **Logistic Regression** performs well right from the start. Platt Scaling maintains Accuracy and improves Log-loss and the Brier score. Isotonic Regression achieves the highest accuracy and the lowest Brier score, but significantly worse Log-loss.
In **Random Forest**, Platt Scaling significantly reduces the log-loss, with accuracy remaining unchanged and the Brier score remaining almost the same. Isotonic Regression, on the other hand, performs worse on all metrics. The reliability diagrams show different results for the two models. Overall, Platt Scaling proves to be more stable, and Isotonic Regression tends to produce more extreme probabilities.

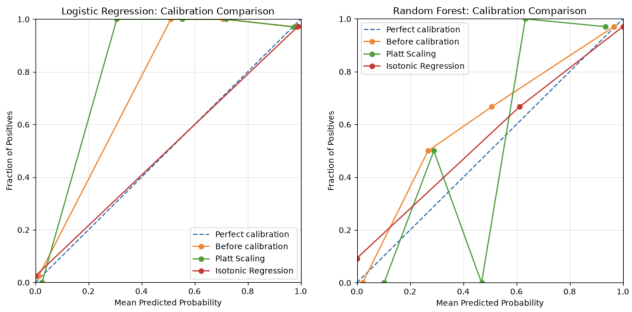

Figure 5.1: Reliability diagrams

5.2 Diabetes Dataset

Model	Calibration	Accuracy	Log-loss	Brier
Logistic Regression	None	**0.7468**	0.5100	0.1687
	Platt	0.7403	**0.5032**	**0.1664**
	Isotonic	0.7403	1.3689	0.1759
Random Forest	None	0.7208	0.5470	0.1863
	Platt	0.7078	**0.5426**	0.1818
	Isotonic	**0.7273**	0.7398	**0.1816**

Table 5.2: Results on the Diabetes dataset

On the Diabetes dataset, for **Logistic Regression**, Platt Scaling slightly reduces the log-loss and Brier score, but with a decrease in accuracy. Isotonic Regression maintains the same accuracy as Platt, but worsens the log-loss.
For **Random Forest**, Platt Scaling improves both the log-loss and the Brier score, but slightly reduces accuracy. Isotonic Regression achieves the highest accuracy and the lowest Brier score, but worsens the log-loss.
The reliability diagrams show different results for the two models: Platt provides the most consistent improvement, while Isotonic is less stable results.

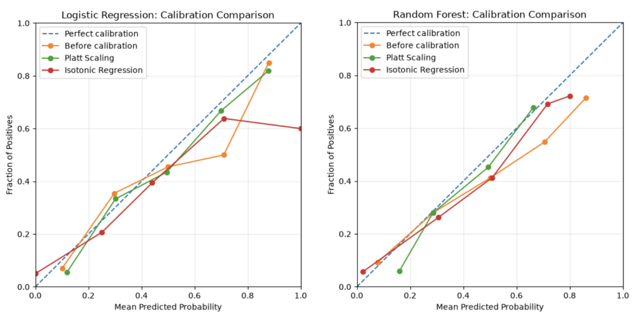

Figure 5.2: Reliability diagrams

Chapter 6

Conclusion

The results of the project show that the effect of calibration depends on the classifier and the dataset. In particular, an improvement in probabilities does not always correspond to an increase in accuracy. Probabilities may improve when the classification performance remains unchanged or decreases slightly.

Among the methods analysed, **Platt Scaling** performs best. In fact, it reduces the log-loss for both classifiers on the Breast Cancer dataset and produces more modest improvements on the Diabetes dataset. The benefits are limited when the original probabilities are already of good quality, as is the case with Logistic Regression on the Breast Cancer dataset.

Isotonic Regression, on the other hand, produces less clear results. In fact, in some cases it improves accuracy or the Brier score, but it can significantly worsen the log-loss. This is because, as mentioned earlier, the method's greater flexibility can favour overfitting when the calibration set is small [1], like in this case.

The relatively small size of the datasets and calibration sets therefore represents a limitation of the analysis. Overall, the experiments confirm that good accuracy does not guarantee well-calibrated probabilities and that post-hoc calibration can improve their accuracy. In the cases analysed, Platt Scaling is the most robust method, whilst Isotonic Regression shows greater sensitivity to the quantity of data available.

Bibliography

[1] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*, 2005.

[2] Maja Pavlovic. Understanding model calibration: A gentle introduction and visual exploration of calibration and the expected calibration error (ece). *ICLR Blogposts*, 2025. arXiv:2501.19047.